Blender: Simulate the Solar System

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Background: This project focuses on modeling planetary motion under gravity, known as Kepler's problem, where the system is simplified into a two-body interaction between the sun and a planet. Since the sun is much more massive, it is treated as fixed at the origin, and the planet moves under a central gravitational force governed by Newton's laws. The resulting equations of motion are expressed in Cartesian coordinates and solved using numerical time-stepping methods to simulate the planet's trajectory. And, also, to further research deep into the topic, the gravitational force between each planet will also be considered for the second part of this toolbox. The new software, Blender, will be introduced through this project and explained how it may change the process of physics simulation.

Description: The computational approach simulates planetary motion by numerically solving the equations of motion derived from Newton's law of gravitation. The Kepler's is deducted as:

$$F_x = -\frac{GMm}{r^2}cos\theta = -\frac{GMm}{r^3}x, F_y = -\frac{GMm}{r^2}sin\theta = -\frac{GMm}{r^3}y$$

Equation: Kepler's law in Cartesian components

And thus, the equation of motion becomes:

$$\frac{d^2x}{dt^2} = -\frac{GM}{r^3}x, \frac{d^2y}{dt^2} = -\frac{GM}{r^3}y$$

Equation: Equation of motion in Cartesian components

The resulting equations are expressed in Cartesian coordinates, yielding a system of coupled second-order differential equations for the x and y components of motion. These equations are then integrated over time using the Verlet integration method, which provides improved numerical stability and accuracy compared to simpler methods. Through this iterative time-stepping process, the simulation approximates the planet's trajectory and captures the characteristics of orbital motion.

Applications: Blender is widely used not only for 3D modeling and animation, but also as a platform for physics-based simulation and visualization. In physics, it can be applied to model dynamical systems such as planetary motion, particle interactions, and fluid behavior, providing an intuitive way to observe complex physical processes. Its scripting capabilities, particularly through Python, allow users to define custom motion based on mathematical models and numerical results, ensuring that objects in the scene evolve according to physically meaningful equations. This makes Blender a valuable tool for prototyping simulations, visualizing computational data, and supporting education by bridging theoretical physics and visual representation. Additionally, techniques from computer graphics and game development can be leveraged to enhance simulation efficiency and interactivity, offering potential for further innovation in scientific visualization.

References:

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